

Unit 1: Electrostatics

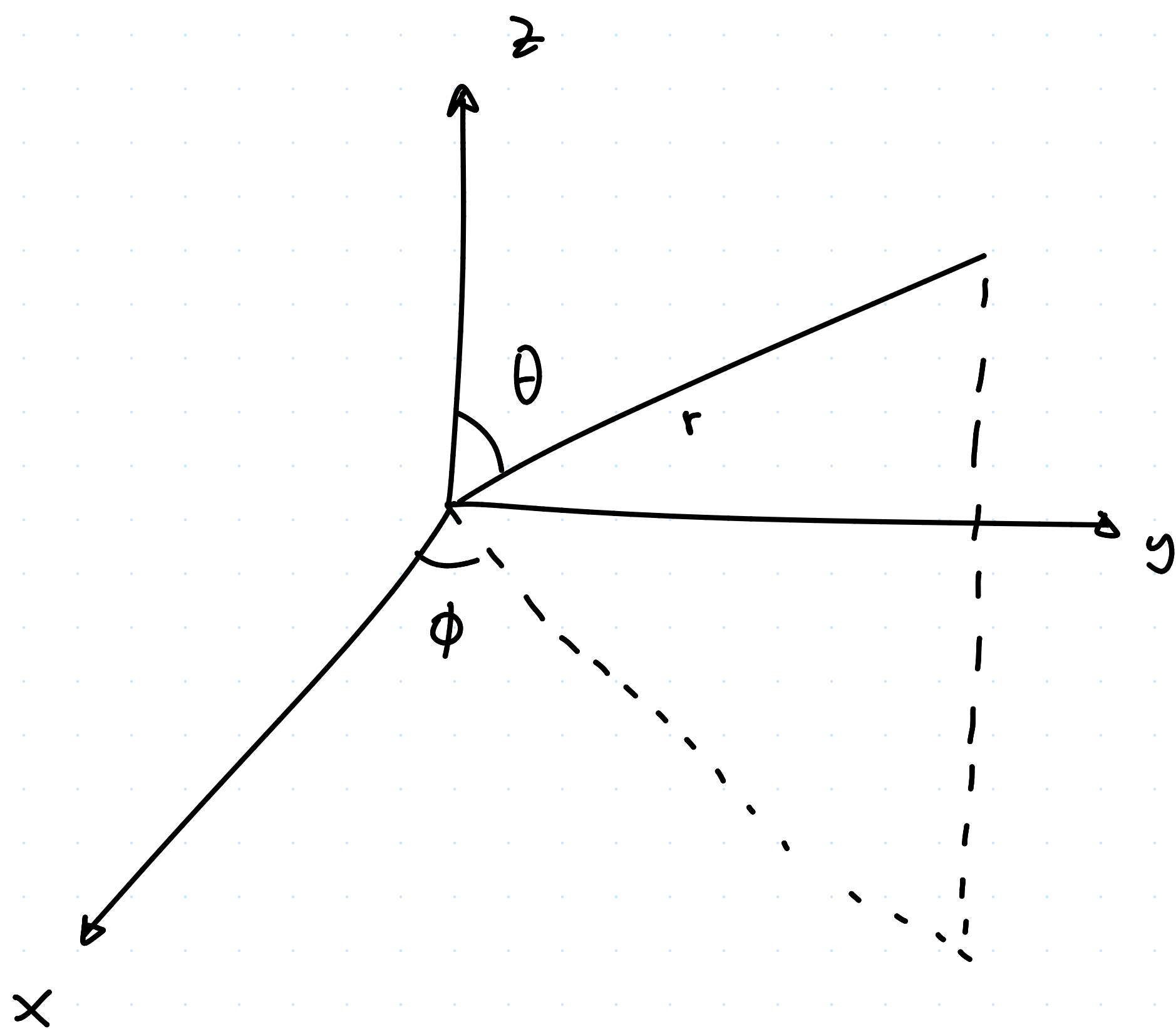
▮ Laplace equation

$$\nabla^2 V = 0$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

▮ solution of Laplace equation

$$V = V(r, \theta)$$



▮ conducting sphere in an electric field.

$$U(r, \theta) = \sum_{k=0}^{\infty} \left[A_k r^k + C_k r^{-(k+1)} \right] P_k \cos \theta$$

Tenemos

$$U(r, \theta) = \sum_{k=0}^{\infty} \left[A_{k+1} r^k + C_{k+1} r^{-(k+1)} \right] P_k \cos \theta$$

Tenemos...

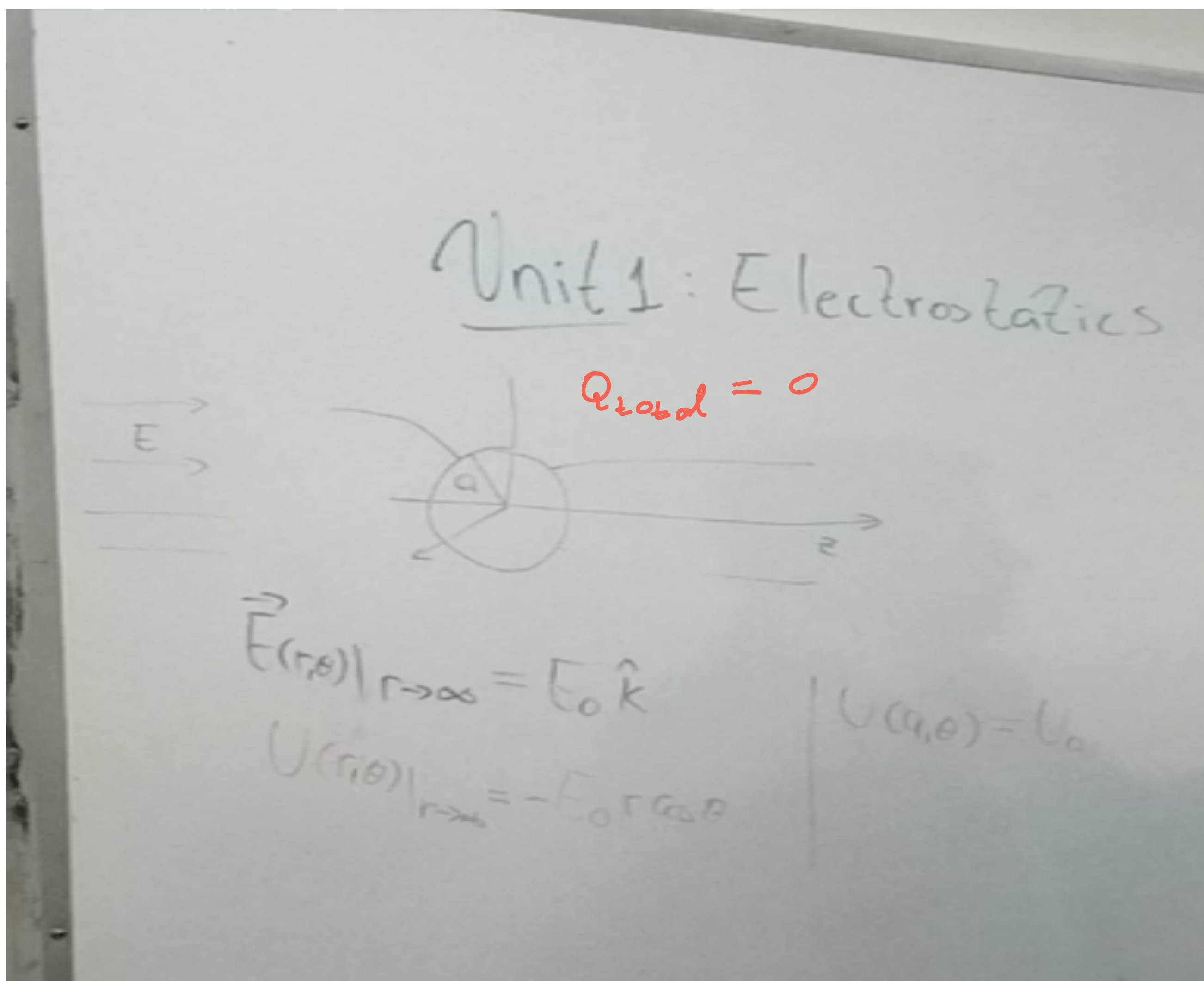
$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) + k P = 0$$

$$U(r, \theta) = A_1 + C_1 r^{-1} + A_2 r \cos \theta + C_2 r^{-2} \cos \theta + A_3 r^2 P_2 \cos \theta \\ + C_3 r^{-3} P_2 \cos \theta + A_4 r^3 P_3 \cos \theta \\ + C_4 r^{-4} P_3 \cos \theta$$

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2} (3x^2 - 1)$$



$$A_2 = -E_0$$

$$A_i = 0$$

$$\underline{i \geq 3}$$

$$C_1 = 0$$

Nos queda:

$$U(r, \theta) = A_1 - E_0 r \cos \theta + C_2 r^{-2} \cos \theta + C_3 r^{-3} P_2 \cos \theta + C_4 r^{-4} P_3 \cos \theta + \dots$$

$$\hat{i}, \hat{j}, \hat{k} \Rightarrow a\hat{i} + b\hat{j} + c\hat{k} = \vec{0} \quad \left| \begin{array}{l} (a-b)\hat{i} + c\hat{j} + d\hat{k} = \vec{0} \\ a-b=0 \Rightarrow a=b \end{array} \right.$$

$$\Rightarrow a=0; b=0; c=0$$

Nuevo conjunto de vectores.

$$\{1, x, x^2, x^3\} \Rightarrow 2 + 3x + x^2 - ex^3$$

$$\left| \begin{array}{l} a + bx + cx^2 + dx^3 = 0 \\ a = b = c = d = 0 \end{array} \right.$$

Es base $\rightarrow$

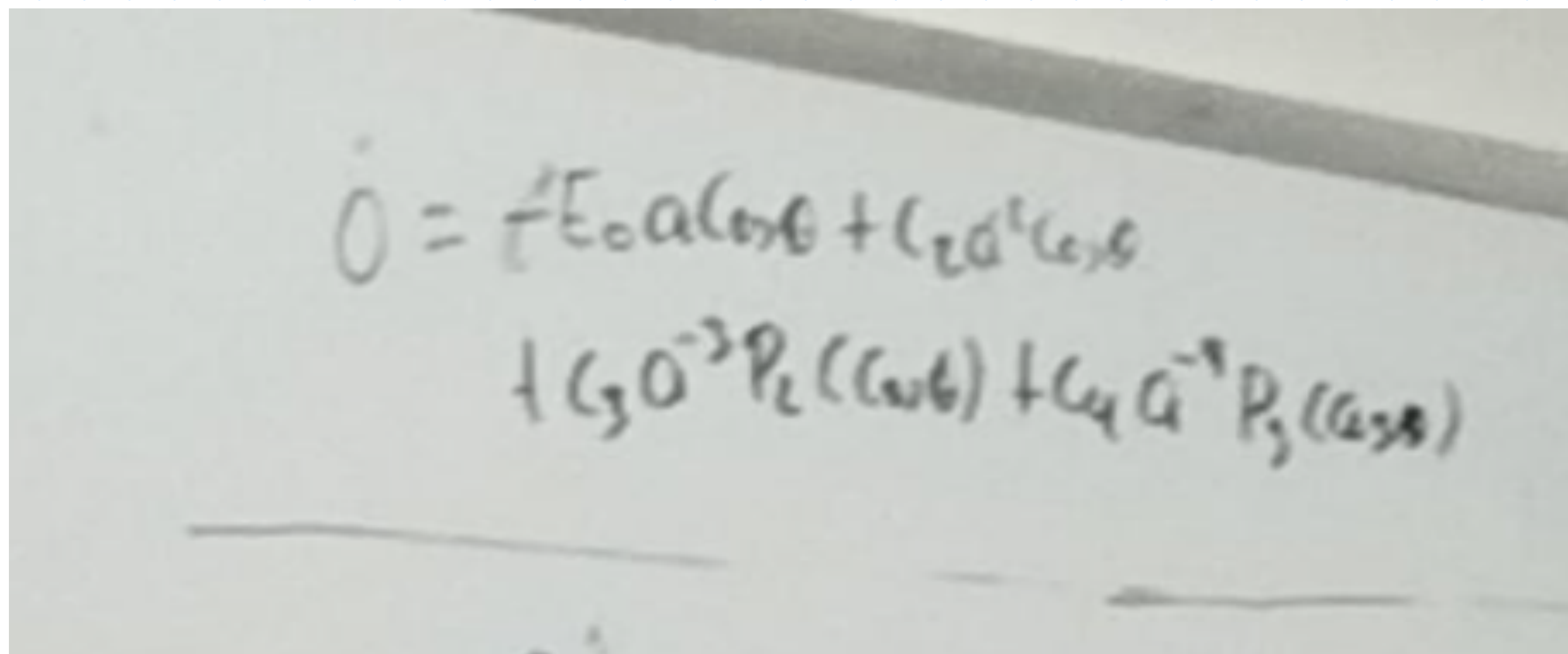
$$\{P_0 \cos \theta, P_1 \cos \theta, \dots, \}$$

$$U(a, \theta) = U_0 = A_1 - \underbrace{E_0 a \cos \theta}_{\cos \theta} + \underbrace{C_2 a^{-2} \cos \theta}_{\cos \theta} + C_3 a^{-3} P_2 \cos \theta + C_4 a^{-4} P_3$$

Deben cancelarse para evitar la dependencia de θ

$$E_0 a \cos \theta = C_2 a^{-2} \cos \theta \quad ; \quad C_i = 0 \quad i \geq 3$$

$$0 = -E_0 a \cos \theta + C_2 a^{-2} \cos \theta + C_3 a^{-3} P_2 \cos \theta + C_4 a^{-4} P_3 \cos \theta$$



$$0 = -E_0 a \cos \theta + C_2 a^{-2} \cos \theta + C_3 a^{-3} P_2(\cos \theta) + C_4 a^{-4} P_3(\cos \theta)$$

Tenemos (Potencial)

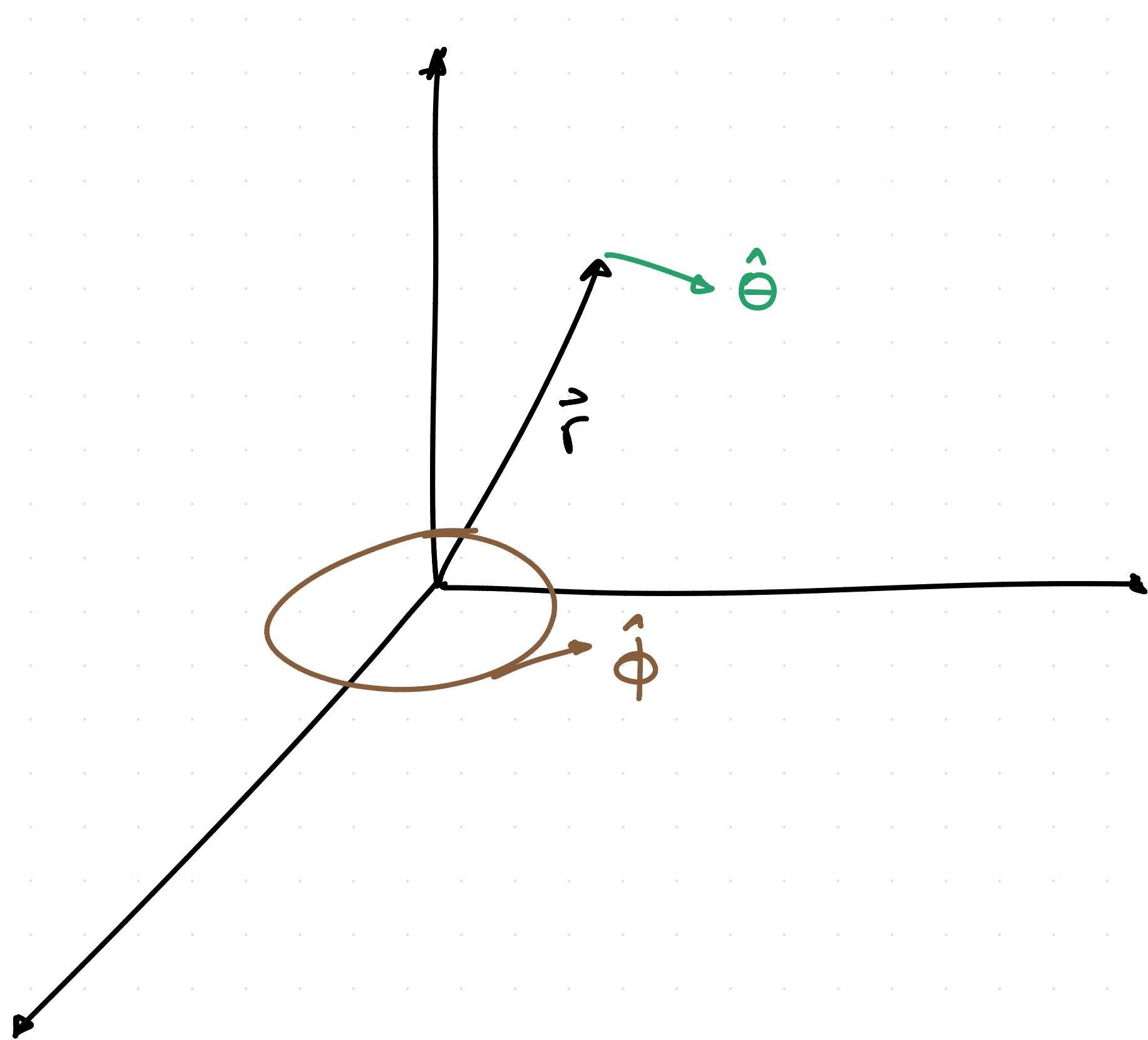
$$U(r, \theta) = U_0 - E_0 r \cos \theta + E_0 a^3 r^{-2} \cos \theta$$

Relación del campo eléctrico con el potencial.

$$-\nabla U = \vec{E}$$

Tenemos

$$\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$



Ejercicio:

Calcular!

$$-\nabla U = \vec{E} = E_r \hat{r} + E_\theta \hat{\theta}$$

Tenemos

$$E_r = - \frac{\partial U(r, \theta)}{\partial r}$$

$$E_\theta = - \frac{1}{r} \frac{\partial U}{\partial \theta}$$

Tenemos (calculando las derivadas parciales)

$$\frac{\partial U}{\partial r} = -E_0 \cos \theta + E_0 a^3 (-2) r^{-3} \cos \theta$$

$$\frac{\partial U}{\partial \theta} = E_0 r \sin \theta - E_0 a^3 r^{-2} \sin \theta.$$



$$U(r, \theta) = U_0 - E_0 r \cos \theta + E_0 a^3 r^{-3} \cos \theta$$

$$-\nabla U = \vec{E} = E_r \hat{r} + E_\theta \hat{\theta}$$

$$\nabla = \frac{\partial}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \hat{\phi}$$



$$E_r = -\frac{\partial U}{\partial r} = -[-E_0 \cos \theta - 2E_0 a^3 r^{-3} \cos \theta] = E_0 \cos \theta \left[1 + \frac{2a^3}{r^3}\right]$$

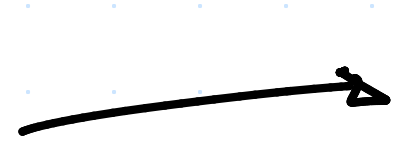
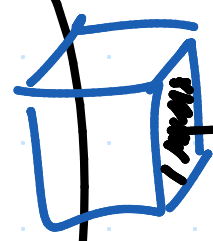
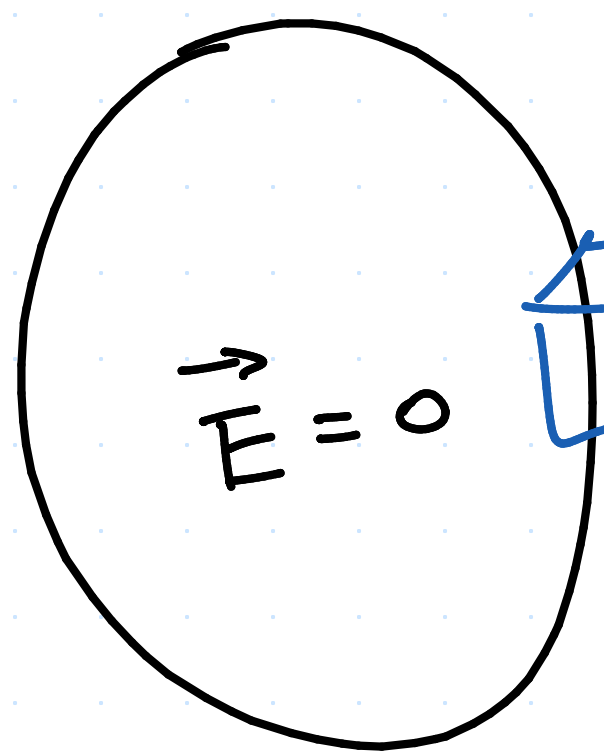
$$E_\theta = -\frac{1}{r} \frac{\partial U}{\partial \theta} = -\frac{1}{r} [E_0 r \sin \theta - E_0 a^3 r^{-2} \sin \theta] = -E_0 \sin \theta \left[1 - \frac{a^3}{r^3}\right]$$

Tenemos

$$E_r = -\frac{\partial}{\partial r} U(r, \theta) = -\left[-E_0 \cos \theta - 2E_0 a^3 r^{-3} \cos \theta\right] = E_0 \cos \theta \left[1 + \frac{2a^3}{r^3}\right]$$

$$E_\theta = -\frac{1}{r} \frac{\partial U}{\partial \theta} = -\frac{1}{r} [E_0 r \sin \theta - E_0 a^3 r^{-2} \sin \theta] = -E_0 \sin \theta \left[1 - \frac{a^3}{r^3}\right]$$

$\sigma(\theta)$



Integramos

$$\vec{E} = E \hat{n}$$

$$0 = \int \sigma(\theta) da$$



muy cerca

de la superficie

$$\vec{E} = E_r \hat{r}$$

Tenemos

$$\vec{E}_r A = \frac{\sigma A}{\epsilon_0}$$

$$\vec{E}_r \epsilon_0 = \sigma$$

$$\sigma(\theta) = 3 \epsilon_0 E_0 \cos \theta$$

Caso electrostático $\rightarrow$ carga en la superficie.

$$Q_{\text{total}} = \int_0^{2\pi} \int_0^{\pi} \sigma a^2 \sin \theta d\theta d\phi$$

$$= \int_0^{2\pi} \int_0^{\pi} 3 \epsilon_0 E_0 \cos \theta a^2 \sin \theta d\theta d\phi = 0$$